

Growth Engine

A plain-English guide to running your automated outreach machine: how to start it, how to open it from your phone, what every screen does, and how to collect and review prospect clients without a single email being sent.

The one thing to understand first

The engine has three modes. It starts in **DRY_RUN**, which means it does **everything except actually send email**. It finds companies, researches them, and writes the emails, then saves each one as a file on your computer for you to read.

You can use it for weeks in **DRY_RUN** with zero risk. Nothing reaches a real person until you deliberately change the mode and confirm it a second time in Telegram.

What this machine does

It replaces the manual work of a salesperson. Every night and every day, on your own computer, it can:

- **Find** real businesses that match who you sell to, from public government license records or other sources.
- **Enrich** them by visiting their website to find an email address and the owner's name.
- **Verify** that each email address actually exists, so your messages do not bounce.
- **Research** each company to find one genuine, specific detail worth mentioning.
- **Write** a short personalized email using that detail, with a local AI model running on your own machine (free, private).
- **Wait for your approval** on every single email, from your phone or the dashboard.
- **Send** slowly and safely, following strict rules that protect your email reputation, then follow up on days 3, 6 and 10 and stop the moment someone replies.
- **Watch replies**, sort them (interested / question / not now / unsubscribe), and alert you with a suggested answer.

Words you will see, in plain English

Prospect — a company that might become a client.

Touch — one email in a sequence (first email, follow-up 1, and so on).

Draft — an email the AI wrote that is waiting for your approval.

Suppression list — people who must never be contacted again. Permanent.

Campaign — your details: company name, what you sell, your links, your signature.

1. Starting the engine

Open the project folder and **double-click** `start.bat`. That is the whole procedure. The first time you run it, it spends a few minutes installing what it needs; after that it starts in a few seconds.

Why not just "python run.py"?

Because that often uses the wrong Python installation on Windows and fails with an error like **No module named 'aiogram'**. `start.bat` always picks the right one for you.

If you prefer typing commands, the correct one is: `.venv\Scripts\python.exe run.py`

A black window opens and stays open. That window **is** the engine — leave it running. When you see this, everything is healthy:

```
=====
Growth Engine v2.0.0
Mode: DRY_RUN (nothing will be sent, .eml files go to /outbox)
Dashboard: http://127.0.0.1:8080
=====
```

Now open **`http://localhost:8080`** in your browser and log in with the password from your `.env` file (the line starting with **`DASHBOARD_PASSWORD=`**).

To stop the engine: click the black window and press **Ctrl + C**, or just close it.

2. Opening the dashboard from your phone

Double-click **`start-with-phone-access.bat`** instead. It does the same thing, plus it creates a free, private web address you can open from anywhere in the world. No router setup, no account, no cost.

A few seconds after starting, the window prints your link:

```
-----
PHONE ACCESS (Cloudflare tunnel): https://something-random.trycloudflare.com
Open that link on your phone and log in with DASHBOARD_PASSWORD.
-----
```

Open that link on your phone, log in with the same password, and you have the full dashboard in your pocket. Tip: use your phone browser's "Add to Home Screen" so it behaves like an app.

Important things about the phone link

The link changes every time you restart the engine. Copy the fresh one each time (or message it to yourself).

Your PC must stay on and running the engine. The link is just a doorway to your computer; if the engine stops, the link stops working.

The link is public but unguessable. Your password is the only lock on the door, so use a strong one. The engine refuses to create a link at all while the password is still the default.

3. Collecting prospects with NO emails sent

This is the safe workflow to use from day one. Everything below happens in DRY_RUN, where sending is physically switched off.

Step 1 — Tell the engine who you are (once)

In the dashboard go to **Admin** → **Campaign** and fill in your company name, one or two sentences about what you sell, your name, your website, your demo link, your booking link, and your signature. Click **Save campaign**.

Until you do this, every email says "Example Co" and an orange banner reminds you.

Step 2 — Find companies

From Telegram, send your bot a message like:

```
/find hvac tampa fl 20
```

That means: find 20 air-conditioning companies in Tampa, Florida. Swap in your own trade and city. The engine replies with progress as it works.

No Telegram bot yet? Do exactly the same thing by typing this in a terminal in the project folder:

```
.env\Scripts\python.exe -m engine.cli find "hvac" "tampa fl" 20
```

Behind the scenes it pulls from official state contractor licence records (public information, no scraping), finds each company's website, reads it for an email address and the owner's name, then checks that the address is real.

Step 3 — Look at what it found

Open the **Leads** page. You get a searchable table: company, trade, city, email, verification level, review count, and status. Click any row for the full profile — the research card, every email in the thread, and what is scheduled next.

Verification levels, simply put:

Level	What it means
SMTP_OK	Best. The mail server confirmed this exact address exists.
MX	Good. The company's mail system is real and reachable.
SYNTAX	Weak. The address looks valid but could not be confirmed.
FAILED	Bad. The engine will never send here. It switches to their contact form instead.

Step 4 — Let the AI write the emails

Send **/draft** in Telegram (or run `... engine.cli draft`). The AI writes one personalized email per verified company. Each arrives on your phone with three buttons:

- **Approve** — good to go (it still will not send in DRY_RUN).
- **Edit** — reply with your corrected wording and it replaces the draft.

- **Skip** — throw this one away.

You can do the same on the **Outbox** page in the dashboard.

Step 5 — Read the emails as real files

Send **/send**. In DRY_RUN this writes each approved email into the **outbox** folder as a **.eml** file. Double-click any of them — they open in your mail app exactly as the recipient would see them, including the unsubscribe footer. Nothing was transmitted.

How to be certain nothing was sent

The header of the dashboard always shows a **DRY_RUN** badge.

Emails appear as files in the **outbox** folder instead of leaving your computer.

The **Activity** page logs every action, and in DRY_RUN each one reads "DRY_RUN: wroteeml".

4. Every screen, explained

Page	What you use it for
Overview	Your scoreboard: totals for found, verified, drafts waiting, sent, replies, interested. The funnel chart shows where prospects drop off; the bounce gauge must stay in the green.
Leads	The full list of companies. Filter by status, trade, city or verification, or search by name. Click a row for everything known about that company.
Outbox	Your approval desk. Drafts waiting for you, approved emails waiting to go, and the schedule of upcoming follow-ups.
Activity	A diary of every action the engine took, newest first. This is where you look when you wonder "why did it do that?"
Settings	Read-only summary of the safety limits, current mode and AI models.
Admin	Where you configure everything. Eight tabs, described below.

5. The Admin tabs

Tab	What it does
Connections	Save your Telegram bot and your email account (multiple allowed, one active at a time). Each has a Test button that really tries to log in and tells you plainly if it worked.
Models	Choose which AI writes, which sorts replies, and which researches. Your local free models are listed automatically. You can also add a paid API (OpenAI, Claude, Groq and others) — there is a daily spending cap and it falls back to your free local model automatically when reached.
Campaign	Who you are and what you sell. Everything in every email comes from here.
Rails	The sending safety limits. You can make them stricter , never looser — the engine rejects any attempt to loosen them.
Templates	Edit the wording of the first email and the three follow-ups, with a live preview against a sample company and automatic style checking.
Prospector	Where prospects come from, which state's records to use, review-count limits, and a blocklist of big franchise names to ignore.
Suppression	The permanent do-not-contact list. You can add and export, but there is deliberately no delete button — respecting an opt-out forever is the point.
Data	Export your prospects, emails and replies to spreadsheets, import a CSV list, or clear out DRY_RUN test files.

6. Telegram commands

Command	What it does
/find hvac tampa fl 20	Find, enrich and verify 20 companies of that trade and city.
/draft	Write a personalized email for every verified company, sent to you for approval.
/send	Release approved emails into the schedule (files only, in DRY_RUN).
/status	A one-message summary of your whole funnel.
/report	The full daily report, on demand.
/models	Which AI is doing which job, and today's API usage.
/pause all	Stop all sending immediately.
/resume all	Resume sending.
/golive	The second key required to enable real sending (see next page).

Only your own Telegram account can command the bot. Everyone else is ignored.

7. The safety rules that protect you

These are built into the program's code. They cannot be raised from any screen — only made stricter.

- **Slow warm-up:** at most 10 emails a day in week one, 20 in week two, 30 after that. Sending too much too early is what gets an address blacklisted.
- **Business hours only:** 9:00 to 16:30 US Eastern, Monday to Friday, with a random 3 to 12 minute gap between emails so it never looks like a machine.
- **Every email carries an unsubscribe line and your postal address**, which is what the US CAN-SPAM law requires.
- **"STOP" means forever.** Anyone who asks out is added to the permanent suppression list and checked before every future send.
- **Automatic emergency brake:** if more than 3% of recent emails bounce, all sending pauses by itself and you get a Telegram alert.
- **Follow-ups cancel instantly** when someone replies, so nobody gets a "just following up" after they already answered.

8. When you are ready to send for real

Do these in order. Never skip step 2 — it is what keeps your emails out of spam folders.

- 1 Run in **DRY_RUN** until you are happy with the wording and the companies being found.
- 2 Set up **SPF, DKIM and DMARC** for your sending domain (your email provider's help pages walk you through it; it is copying three DNS records). Use a separate domain from your main company one.
- 3 Switch to **SANDBOX**: real sending, but every email goes only to your own inbox. Run it for about three days and confirm everything looks right.
- 4 Switch to **LIVE**, restart, then send **/golive** in Telegram and confirm. Sending now begins at 10 emails a day.

Two keys are always required for LIVE

Changing the mode is done by editing the `.env` file (**ENGINE_MODE=LIVE**) and restarting — deliberately not a button in the dashboard.

Then you must confirm once more with **/golive** in Telegram. Going live can never happen by one accidental click.

9. If something goes wrong

Symptom	What to do
"No module named ..." when starting	You used the wrong Python. Double-click start.bat instead.
Dashboard will not open	Check the black engine window is still running, and use http://localhost:8080 (not https).

Symptom	What to do
Phone link does not work	It changes on every restart — copy the newest one from the window. Also check your PC is awake and the engine is running.
Drafts come out generic or empty	The local AI is not running. Open a terminal and run ollama list ; start Ollama, then try /draft again.
Prospects found, but no emails	Normal. Many small companies hide their address; those switch to the contact-form path and keep the owner name and licence number for you.
I want everything to stop, now	Send /pause all in Telegram, or close the engine window.

Every action is written to the **Activity** page and to the **Logs** folder, so nothing the engine does is a mystery.

Growth Engine v2.0.0 — you own the machine, the data, and the AI. Nothing leaves your computer unless you approve it.